Sign Selection Preservation under Bounded Score Perturbations

Research Architecture Runtime

`operator/sign`

2026-07-29

Abstract

For $\mathrm{sign}(x) \in \{+1, -1, 0\}$, if $|x' - x| \leq \varepsilon$ then $\varepsilon < x \Rightarrow \mathrm{sign}(x') = +1$, $x < -\varepsilon \Rightarrow \mathrm{sign}(x') = -1$, and $(\varepsilon, x) = (0, 0) \Rightarrow \mathrm{sign}(x') = 0$. Lean `LEAN_FULL`.

This theorem is: `Primitive`

1 Problem

The sign operator returns $\mathrm{sign}(x) = +1$ if $x > 0$, -1 if $x < 0$, and 0 if $x = 0$.

2 Stability notion

Perturbation: $|x' - x| \leq \varepsilon$. Preservation uses *strict* buffers $\varepsilon < x$ and $x < -\varepsilon$. Zero output is preserved only when $\varepsilon = 0$ and $x = 0$. Sharpness covers $(0, \varepsilon]$, $[-\varepsilon, 0)$, and $x = 0$ with $\varepsilon > 0$.

3 Definitions

Definition 1 (sign)**.** $\mathrm{sign}(x) = 1$ if $0 < x$; -1 if $x < 0$; else 0.

4 Theorem

Theorem 2 (Sign preservation)**.** *If $|x' - x| \leq \varepsilon$, then: (i) $\varepsilon < x \Rightarrow \mathrm{sign}(x') = 1$; (ii) $x < -\varepsilon \Rightarrow \mathrm{sign}(x') = -1$; (iii) $\varepsilon = 0 \wedge x = 0 \Rightarrow \mathrm{sign}(x') = 0$.*

Theorem 3 (Sharpness)**.** *If $0 < x \leq \varepsilon$ then some $|x' - x| \leq \varepsilon$ has $\mathrm{sign}(x') \neq 1$; symmetrically for $-\varepsilon \leq x < 0$; if $x = 0$ and $\varepsilon > 0$ then some $|x' - x| \leq \varepsilon$ has $\mathrm{sign}(x') \neq 0$.*

5 Intuition

A strict buffer larger than ε keeps the perturbed point away from 0. On the closed band the push $x \mapsto x - \varepsilon$ can leave the positive ray.

6 Examples

Example 4. $x = 2$, $\varepsilon = 0.5$: $\varepsilon < x$, so the sign stays $+1$. $x = 0.4$, $\varepsilon = 0.5$: $x' = x - \varepsilon = -0.1$ has sign -1.

7 Proof

From $|x' - x| \leq \varepsilon$, $x - \varepsilon \leq x' \leq x + \varepsilon$. If $\varepsilon < x$ then $x' > 0$, so $\operatorname{sign}(x') = 1$. If $x < -\varepsilon$ then $x' < 0$, so $\operatorname{sign}(x') = -1$. If $\varepsilon = 0$ and $x = 0$ then $x' = 0$. This is Theorem 2.

Sharpness: if $0 < x \leq \varepsilon$, take $x' = x - \varepsilon \leq 0$. If $-\varepsilon \leq x < 0$, take $x' = x + \varepsilon \geq 0$. If $x = 0$ and $\varepsilon > 0$, take $x' = \varepsilon$. This is Theorem 3.

8 Formal statement

Lean module
`Research.Operators.Sign.Preservation`

Source file
`lean/Research/Operators/Sign/Preservation.lean`

Theorems

- `sign\_preservation`
- `sign\_sharpness`

Certificate
`lean/certificates/sign/sign-preservation`

Status
`LEAN_FULL` on Mathlib $\mathbb{R}$ (`REAL_MATHLIB`)

9 Proof dependencies

- Absolute-value ball arithmetic on $\mathbb{R}$.
- No other operator theorems.

10 Consequences

Primitive trichotomy gate; abs-threshold and interval membership use analogous 1D buffers.